Biobehavioral clocks predict dementia and survival across Latin American populations

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Abstract

Behavioral and health-related measures provide critical markers of accelerated aging, yet their links with dementia and survival remain unclear. Here we tested whether biobehavioral age gaps (BBAGs), the difference between chronological age and predicted age derived from protective and risk factors, capture pathological aging trajectories across Latin America. We analyzed harmonized cross-sectional ($n = 12,693$) and longitudinal ($n = 8,391$) data from six Latin American countries. Age prediction models showed robust performance across progressively stricter specifications excluding dementia-related cognitive, mood, and functional predictors (variance explained from $R^2 = 0.21$ to 0.30). Individuals with dementia showed higher BBAGs across different diagnostic definitions, with large effect sizes in the full model (δ up to 0.81, $P < 1 \times 10^{-15}$) that remained significant after excluding predictors related to diagnosis ($\delta = 0.42$ to 0.57, all $P < 1 \times 10^{-15}$). Higher BBAGs were associated with increased mortality risk, with each standard deviation increase conferring a 41–48% higher hazard after adjustment for age, sex, and education (hazard ratios 1.41–1.48, all $P < 1 \times 10^{-15}$). Baseline BBAGs predicted future accelerated aging at follow-up ($r = 0.38$–0.47, all $P < 1 \times 10^{-15}$), indicating longitudinal stability and prognostic value**.** Findings were consistent across countries, diagnostic strata, and sensitivity analyses, with country-level meta-analyses confirming pooled

mortality effects with low heterogeneity (HR ≈ 1.60, 95% CI 1.53–1.68; $I^2 = 31.5\%$). These results identify BBAGs as scalable and ecological markers of systemic aging vulnerability linked to dementia and survival, with particular relevance for early detection and intervention in underrepresented settings.

1. Introduction

Aging is shaped by behavioral and whole-body health-related factors that capture how organisms function in real-world contexts[1-5]. Measures of behavior, cognition, functionality, and health provide a rich, noninvasive readout of an organism's integrated state, capturing action, physiology, and accumulated health burden over time[6-13]. Animal research demonstrates that behavior can track aging trajectories and biological mechanisms[2,14-17]. Lifelong biobehavioral profiling in vertebrates has distinguished individuals with shorter or longer lifespan and predicted chronological age, omics profiles, and lifespan[2]. In humans, biobehavioral age gaps (BBAGs)[4,18] quantify the difference between chronological age and age estimated via aging clocks trained on behavioral and health-related protective and risk factors. Large-scale evidence across 40 countries shows that BBAGs capture deviations from normative aging trajectories and predict later cognitive and functional decline as well as future accelerated aging[4]. This approach provides scalable aging-clock metrics based on behavioral and health variables that are widely accessible at the population level. Because behavioral profiles reflect whole-body health, they are closely linked to brain health and dementia risk[5,18-23]. Therefore, biobehavioral clocks may serve as ecologically valid and cost-effective indicators of aging trajectories and as early markers of accelerated aging, potentially indexing dementia risk and survival.

Despite this progress, there is no evidence linking BBAGs to dementia status nor to early mortality, the most clinically consequential outcomes of aging[24-26]. Animal models cannot be translated directly to human dementia phenotypes[27], and BBAG studies in humans have focused on population-based healthy aging rather than clinical outcomes[28,29]. Prior work also left unresolved whether BBAG effects are independent of variables closely tied to dementia, as some predictors used to estimate them (e.g., cognition, functionality) may conceptually overlap with diagnosis[30,31]. This raises a critical question as to whether BBAGs capture independent aging-related processes or primarily reflect clinical features used to define disease, particularly dementia. Studies that progressively remove variables overlapping with dementia diagnosis (such as cognitive or functional scores) are still not systematically implemented across dementia research, despite their importance for reducing circularity when evaluating associations with clinical outcomes. It also remains unknown whether BBAGs predict survival and future accelerated aging, especially in cohorts with increased dementia risk. Finally, evidence from underrepresented and diverse regions remains limited, particularly in Latin American and Caribbean populations, where dementia burden is high, behavioral health impairments are more pronounced than in high-income populations, and access to biological biomarkers is often constrained[10,32-35]. Hence, the field still lacks a direct test of whether BBAGs represent markers of pathological aging.

To address these gaps, we examined BBAGs in a large multicountry Latin American cohort using both cross-sectional and longitudinal data (Fig.1). Age was estimated from harmonized biobehavioral factors associated with risk (whole-body, mental, cardiometabolic, neurological, and overall health impairments, as well as disability, and clinical criteria for cognitive impairment and dementia) and protection (physical activity, healthcare engagement, cognitive reserve, economic resources, education, and other sociodemographic variables)[28,29]. . BBAGs were estimated as the difference between predicted age and chronological age, with correction for age-related biases. To test robustness and reduce circularity, three analytic models were used: a full model including all predictors; a model excluding variables directly involved in diagnostic algorithms (cognitive-negative: excluding cognitive scores and related diagnostic information); and a stricter model excluding both cognitive and other clinical variables related to dementia phenotypes (cognitive, mood, functionality). We hypothesized that BBAGs estimated with partially different protective and risk factors would replicate previous findings[4] and that age would be reliably estimated across progressively stricter model definitions. Individuals with dementia would exhibit larger

BBAGs than those without dementia, even after excluding predictors associated with clinical diagnosis. Finally, we anticipated that accelerated BBAGs would predict lower survival and future accelerated aging during follow-up, with effects remaining after adjustment for age, sex, and education across diagnostic strata.

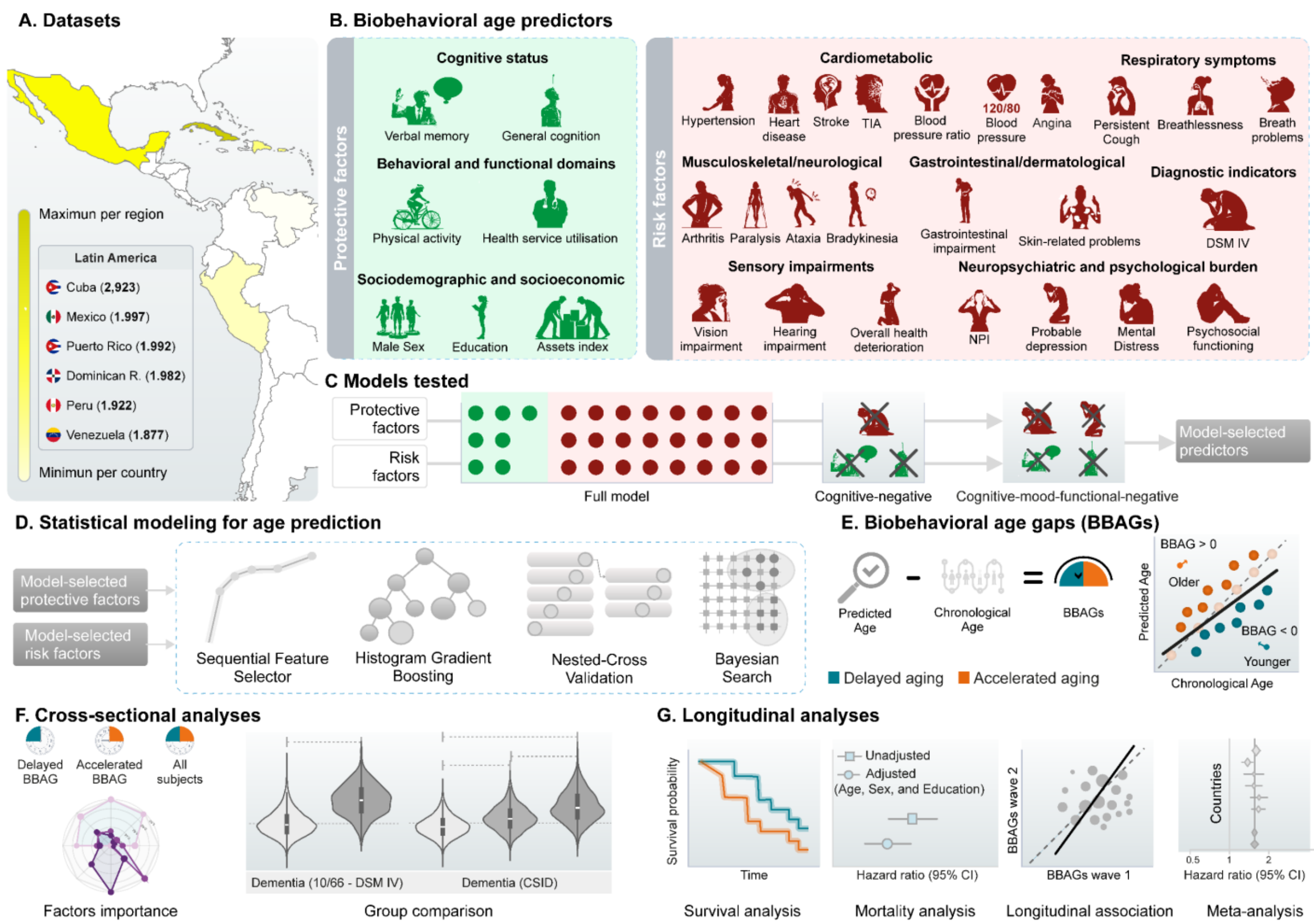


Figure 1. Study design and analytical pipeline. A) Database. Geographical distribution of participants across Latin America, with country-level shading proportional to sample size. **B) Predictors.** Biobehavioral age predictors, grouped into protective (cognitive status, behavioral and functional domains, and sociodemographic and socioeconomic factors) and risk factors (cardiometabolic, respiratory symptoms, musculoskeletal and neurological, gastrointestinal, diagnostic indicators, sensory impairments, and neuropsychiatric and psychological burden). **C) Model construction**. Illustration of the modeling framework, including a full model integrating all predictors and reduced models derived from cognitively negative and mood-functional-negative approaches to identify model-selected features. **D) Statistical approach.** Machine learning pipeline combining sequential feature selection, gradient boosting models, nested cross-validation, and Bayesian optimization. **E) Biobehavioral age gap estimation (BBAGs).** BBAGs were defined as the difference between predicted and chronological age, where positive values indicate accelerated aging and negative values indicate delayed aging. **F) Cross-sectional analyses**. Contributions of protective and risk factors and differences in BBAGs across dementia groups. **G) Longitudinal analyses.** Associations of BBAGs with survival, risk estimates, their evolution over time, and country-level meta-analysis.

2. Results

We analyzed a cross-sectional sample of 12,693 individuals (9.30% dementia) recruited from Latin American countries: Cuba (n=2,923, 10.8% dementia), the Dominican Republic (n=1,982, 11.9% dementia), Mexico (n=1,997, 8.90% dementia), Peru (n=1,922, 8.40% dementia), Puerto Rico (n=1,992, 11.4% dementia), and Venezuela (n=1,877, 5.80% dementia). The longitudinal cohort (n=8,391, 14.4% dementia) comprised individuals from Cuba (n=1,970, 13.2% dementia), the Dominican Republic (n=1,170, 18.3% dementia), Mexico (n=1,455, 14.64% dementia), Peru (n=1,309, 10.92% dementia), Puerto Rico (n=1,238, 15.6% dementia), and Venezuela (n=1,249, 14.7% dementia). Participants were drawn from the Global 10/66 study[36,37], a population-based cohort examining aging and dementia across low- and middle-income countries (Table 1). Predictors captured protective and risk factors of biobehavioral aging. Protective factors included cognitive, behavioral, and sociodemographic domains[4]. Cognitive performance was assessed with CERAD[38] and CSID[39], and behavioral indicators included physical activity and healthcare engagement. Sociodemographic variables included education, assets index, and sex. Risk factors included whole-body and overall health burden (cardiometabolic, respiratory, neurological and motor, gastrointestinal, dermatological, and sensory conditions), clinical criteria (DSM-IV[40] and ICD-10[41]), and markers such as pain, lifetime health burden, functional disability (WHO-DAS II[42]), and family dementia history. Neuropsychiatric burden included NPI[43], Euro-D [44], and SRQ [45] (Details on site harmonization are provided in Supplementary S1). Sequential feature selection (SFS)[46] was used for dimensionality reduction. Age prediction used histogram-based Gradient Boosting[47,48] within nested cross-validation with Bayesian optimization[49]. Three models were trained to reduce diagnostic circularity: Model A (all predictors), Model B (cognitive-negative), and Model C (cognitive-mood-functional-negative). We examined associations with mortality using Cox models[50], compared survival (Kaplan-Meier [51]) across accelerated and delayed aging groups, and assessed longitudinal associations between BBAGs at waves 1 and 2. Meta-analysis validated the consistency of mains effects across countries and models. Sensitivity analyses included models adjusted for age, sex, education, diagnostic stratification, other adjustments, and comparison with mortality-related indexes.

Table 1. Samples demographics

Cross-sectional						
Countries	**Age mean (SD)**	**N (% Female)**	**Dementia (N)**	**Education**		
				College	**Middle/High**	**None/Elementary**
Cuba	75.1 (7.0)	2923 (64.9%)	315	17.0%	58.2%	24.8%
Dominican R.	75.2 (7.5)	1982 (66.0%)	235	3.5%	25.5%	71.0%
Mexico	74.3 (6.6)	1997 (63.2%)	117	5.5%	23.7%	70.8%
Peru	74.8 (7.4)	1922(61.2%)	161	16.8%	64.9%	18.3%
Puerto Rico	76.3 (7.4)	1992 (67.3%)	227	20.6%	56.5%	22.9%
Venezuela	72.3 (6.7)	1877 (63.3%)	108	4.8%	64.4%	30.8%

Longitudinal						
Countries	**Age mean (SD)**	**N (% Female)**	**Dementia (N)**	**Education**		
				College	**Middle/High**	**None/Elementary**
Cuba	78.4 (6.3)	1970 (66.4%)	256	17.3%	59.8%	22.9%
Dominican R.	79.0 (6.7)	1170 (69.1%)	213	4.0%	26.1%	69.9%
Mexico	76.7 (6.2)	1455(64.6%)	213	5.4%	25.1%	69.5%
Peru	77.5 (6.9)	1309(62.4%)	143	16.1%	65.0%	18.9%
Puerto Rico	79.5 (6.5)	1238 (68.3%)	190	22.9%	57.3%	19.8%
Venezuela	76.0 (6.2)	1249 (64.8%)	183	4.9%	67.5%	27.6%

2.1. Chronological age estimation

Chronological age was estimated using biobehavioral predictors under the 3 modeling configurations (Models A–C). Model A achieved the highest performance ($R^2 = 0.30 \pm 0.02$, $f^2 = 0.45$, RMSE = 6.01, MAE = 4.86), followed by Model B ($R^2 = 0.24 \pm 0.02$, $f^2 = 0.34$, RMSE = 6.27, MAE = 5.11) and Model C ($R^2 = 0.21 \pm 0.02$, $f^2 = 0.28$, RMSE = 6.37, MAE = 5.19). In Model A, the most influential predictors (top-5) were general cognition, delayed verbal memory, hearing problems, functional disability, and bradykinesia. In Model B, functional disability, hearing problems, walking, bradykinesia, and education emerged as the strongest predictors. In Model C, the leading predictors included walking, physical activity, hearing problems, bradykinesia, and education (Fig. 2-A1). Selected predictors and their feature importance for each model are provided in Supplementary S2, Table 1. These analyses were replicated in wave 2, yielding similar results (Supplementary S2 Fig. 1-A).

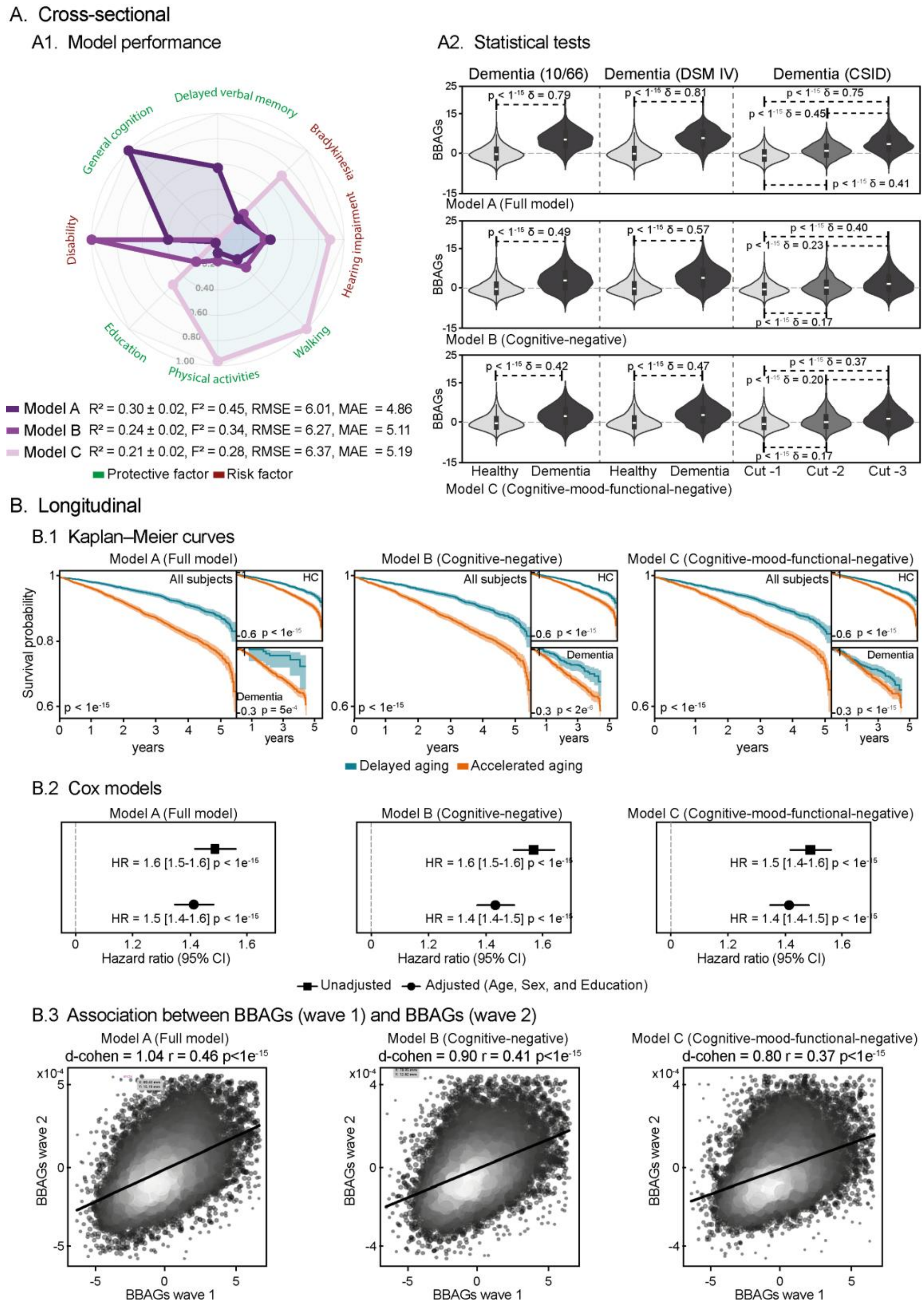


Figure 2. Cross-sectional and longitudinal validation of the biobehavioral age gaps (BBAGs) across model specifications. **(A) Cross-sectional analyses**. (A1) Model performance and feature contributions across three model configurations: Model A (full model), Model B (cognitive-negative), and Model C (cognitive-mood-functional-negative). Radar plots display the relative contribution of top-5 predictors grouped into cognitive, behavioral, and clinical domains, distinguishing protective (green) and risk (red) factors. The complete list of predictors selected for each model, along with their feature importance, is provided in Supplementary S2 Table 1. Model performance metrics (R^2, f^2, RMSE, and MAE) are reported for each configuration. (A2) Distribution of BBAGs across diagnostic groups defined by 10/66, DSM-IV, and CSID criteria. Violin plots show group differences between cognitively healthy individuals and participants with dementia across models,

including additional stratifications (Cut-1, Cut-2, Cut-3). Statistical comparisons (two-sided tests) indicate significant differences across diagnostic categories. **(B) Longitudinal analyses.** (B1) Kaplan–Meier survival curves comparing individuals with accelerated versus delayed aging across all subjects and stratified by diagnostic groups (healthy controls and dementia). Across all model configurations, accelerated aging is associated with reduced survival probability. (B2) Cox proportional hazards models showing the association between BBAGs and mortality risk. Hazard ratios (HR) are reported for unadjusted and adjusted models (age, sex, and education), demonstrating a robust association between higher BBAGs (accelerated aging) and increased mortality risk across all model configurations. (B3) Longitudinal association between BBAGs at wave 1 and wave 2. Scatterplots with fitted regression lines show significant positive associations across models, with effect sizes reported as Cohen's d and Pearson correlation coefficients, indicating temporal stability of the BBAG measure.

2.2. Biobehavioral age gaps predict dementia status

BBAG distributions differed by dementia status under different diagnostic definitions (Fig. 2-A2). Using the 10/66 and DSM-IV criteria, participants classified with dementia consistently showed accelerated aging in all model specifications. Effect sizes were large in Model A (10/66: $\delta = 0.79$; DSM-IV: $\delta = 0.81$; $p < 1\times10^{-15}$ for both), remained large in Model B (10/66: $\delta = 0.49$; DSM-IV: $\delta = 0.57$; $p < 1\times10^{-15}$), and were medium to large in Model C (10/66: $\delta = 0.42$; DSM-IV: $\delta = 0.47$; $p < 1\times10^{-15}$).

Similar patterns were observed when applying CSID classification thresholds. Effect sizes decreased across progressively stricter cutoffs in Model A (Cut −1 vs −2: $\delta = 0.41$; Cut −2 vs −3: $\delta = 0.45$; Cut −1 vs −3: $\delta = 0.75$; all $p < 1\times10^{-15}$), corresponding to medium, medium, and large effects, respectively. In Models B and C, effects were small to medium (Model B: $\delta = 0.17$, 0.23, and 0.40; Model C: $\delta = 0.17$, 0.20, and 0.37). These results were replicated in wave 2 (Supplementary S3 Fig. 1-A).

2.3. Accelerated biobehavioral aging predicts lower survival

Survival analyses revealed lower survival probabilities among individuals with accelerated aging compared with those with delayed aging across all model configurations (Fig. 2-B1). Kaplan–Meier curves showed large group differences in Model A (log-rank $p < 1\times10^{-15}$), Model B (log-rank $p < 1\times10^{-15}$), and Model C (log-rank $p < 1\times10^{-15}$). In unadjusted Cox proportional hazards models, accelerated aging was consistently associated with increased mortality risk across models, with each one standard deviation increase in BBAG conferring a 48–62% higher hazard (Model A: HR = 1.62, 95% CI 1.55–1.70; Model B: HR = 1.57, 1.50–1.64; Model C: HR = 1.49, 1.42–1.56; all $p < 1\times10^{-15}$) (Table 2).

Table 2. Hazard ratios

	Model A		Model B		Model C	
	HR (CI 95%)	p-value	HR (CI 95%)	p-value	HR (CI 95%)	p-value
Unadjusted						
BBAGs (z)	1.62 [1.55-1.70]	$< 1e^{-15}$	1.57 [1.50-1.64]	$< 1e^{-15}$	1.48 [1.42-1.56]	$< 1e^{-15}$
Adjusted						
BBAGs (z)	1.48 [1.41-1.55]	$< 1e^{-15}$	1.44 [1.37-1.50]	$< 1e^{-15}$	1.41 [1.35-1.48]	$< 1e^{-15}$
Age	1.08 [1.08-1.09]	$< 1e^{-15}$	1.08 [1.08-1.09]	$< 1e^{-15}$	1.09 [1.08-1.10]	$< 1e^{-15}$
Sex (Male)	1.51 [1.35-1.70]	$< 1e^{-10}$	1.51 [1.35-1.69]	$< 1e^{-10}$	1.48 [1.32-1.66]	$< 1e^{-10}$
Education	0.98 [0.94-1.03]	0.48	0.99 [0.95-1.05]	0.94	0.99 [0.95-1.05]	0.96

2.4. Baseline BBAGs predict future accelerated aging

Baseline BBAG values were significantly associated with BBAG values at wave 2 across all model configurations (Fig. 2-B3). The strongest association was observed in Model A ($r = 0.47$, Cohen's $d = 1.06$, $p < 1\times10^{-15}$), followed by Model B ($r = 0.41$, Cohen's $d = 0.90$, $p < 1\times10^{-15}$) and Model C ($r = 0.38$, Cohen's $d = 0.82$, $p < 1\times10^{-15}$), corresponding to large effect sizes across all models, indicating that baseline biobehavioral aging profiles significantly predict subsequent accelerated aging patterns.

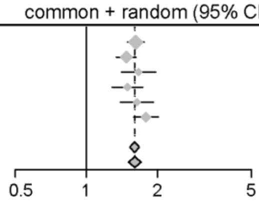
2.5. Associations between BBAGs and mortality across countries and models

Accelerated aging was consistently associated with increased mortality risk across all countries and model specifications. In Model A, pooled estimates showed a hazard ratio (HR) of 1.60 (95% CI 1.53–1.68) under common-effects and 1.60 (1.51–1.71) under random-effects models (Fig. 3-A), with heterogeneity below 50% ($I^2 = 31.5\%$, $p = 0.199$). In Model B, effect sizes remained comparable, with pooled HRs of 1.55 (1.49–1.63) and 1.56 (1.46–1.67) for common- and random-effects models, respectively (Fig. 3-B), with heterogeneity below 50% ($I^2 = 46.8\%$, $p = 0.094$). In Model C, effect sizes were slightly attenuated but remained significant, with pooled HRs of 1.49 (1.42–1.56) under common-effects and 1.49 (1.40–1.58) under random-effects models (Fig. 3-C), with heterogeneity below 50% ($I^2 = 29.7\%$, $p = 0.212$).

Meta-analysis

A. Model A (Full model)

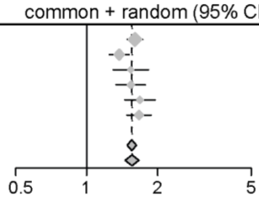
Country	logHR	SE(logHR)	Hazard Ratio common + random (95% CI)	Weight (%) common	Weight (%) random	Hazard Ratio common + random (95% CI)
Cuba	0.48	0.04		34.2	26.5	1.62 [1.49 - 1.76]
Dominican R.	0.39	0.05		24.2	22.3	1.48 [1.34 - 1.62]
Peru	0.51	0.09		8.0	10.6	1.66 [1.41 - 1.97]
Venezuela	0.40	0.08		9.8	12.5	1.49 [1.28 - 1.73]
Mexico	0.49	0.08		8.4	11.1	1.64 [1.39 - 1.92]
Puerto Rico	0.58	0.06		15.3	17.0	1.79 [1.59 - 2.02]
Total common-effects (95% CI), common				100		1.60 [1.53 - 1.68]
Total random-effects (95% CI), random					100	1.60 [1.51 - 1.71]



Heterogeneity: $\tau^2 = 0.002$ $\chi^2 = 7.30$ df = 5 $p < 0.1993$ $I^2 = 31.5\%$

B. Model B (Cognitive-negative)

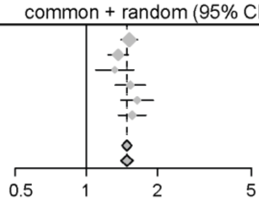
Country	logHR	SE(logHR)	Hazard Ratio common + random (95% CI)	Weight (%) common	Weight (%) random	Hazard Ratio common + random (95% CI)
Cuba	0.47	0.04		36.2	24.7	1.60 [1.49 - 1.73]
Dominican R.	0.32	0.05		22.2	20.7	1.37 [1.25 - 1.51]
Peru	0.43	0.09		7.0	10.7	1.54 [1.30 - 1.83]
Venezuela	0.43	0.07		10.1	13.7	1.54 [1.33 - 1.77]
Mexico	0.52	0.08		9.4	13.1	1.68 [1.45 - 1.95]
Puerto Rico	0.21	0.06		14.9	17.1	1.67 [1.48 - 1.88]
Total common-effects (95% CI), common				100		1.55 [1.49 - 1.63]
Total random-effects (95% CI), random					100	1.56 [1.46 - 1.67]



Heterogeneity: $\tau^2 = 0.003$ $\chi^2 = 9.40$ df = 5 $p < 0.0942$ $I^2 = 46.8\%$

C. Model C (Cognitive-mood-functional-negative)

Country	logHR	SE(logHR)	Hazard Ratio common + random (95% CI)	Weight (%) common	Weight (%) random	Hazard Ratio common + random (95% CI)
Cuba	0.42	0.04		36.7	29.1	1.52 [1.40 - 1.64]
Dominican R.	0.31	0.05		23.5	22.6	1.36 [1.24 - 1.50]
Peru	0.28	0.09		6.7	8.8	1.32 [1.10 - 1.59]
Venezuela	0.43	0.07		10.8	13.0	1.54 [1.33 - 1.78]
Mexico	0.50	0.08		9.3	11.5	1.64 [1.40 - 1.92]
Puerto Rico	0.45	0.07		13.0	15.0	1.56 [1.37 - 1.78]
Total common-effects (95% CI), common				100		1.49 [1.42 - 1.56]
Total random-effects (95% CI), random					100	1.49 [1.40 - 1.58]



Heterogeneity: $\tau^2 = 0.0015$ $\chi^2 = 7.11$ df = 5 $p < 0.2123+$ $I^2 = 29.7\%$

Figure 3. Meta-analysis (countries x models) of Biobehavioral age gaps (BBAGs) and mortality risk. Forest plots show hazard ratios (HRs) and 95% confidence intervals for each country and pooled estimates using both common- and random-effects models across three modeling configurations: Model A (full model), Model B (cognitive-negative), and Model C (cognitive-mood-functional-negative). All models show consistent positive associations between accelerated aging and increased mortality risk, with low-to-moderate heterogeneity across countries.

2.6. Sensitivity analysis

Chronological age prediction in healthy participants: Chronological age was re-estimated using only healthy aging participants and biobehavioral predictors across the three modeling configurations (Models A–C; Supplementary S3 Fig 1-A). Model A achieved the highest performance ($R^2 = 0.24 \pm 0.02$, $f^2 = 0.31$, RMSE = 5.87, MAE = 4.76), followed by Model B ($R^2 = 0.18 \pm 0.02$, $f^2 = 0.26$, RMSE = 6.08, MAE = 4.96) and Model C ($R^2 = 0.17 \pm 0.02$, $f^2 = 0.24$, RMSE = 6.15, MAE = 5.00), corresponding to medium ~~to large effect~~ sizes, with the strongest effects observed in Model A.

Meta-analysis in healthy participants: Accelerated aging in healthy individuals was consistently associated with an increased risk of mortality across all countries and model specifications. Pooled estimates yielded an approximate hazard ratio (HR) of 1.34 (95% CI, 1.26–1.42) under both common- and random-effects models (Supplementary S3 Fig. 1B).

Chronological age prediction in wave 2: Chronological age estimated in the wave 2 (all models, Supplementary S3 Fig 1-C) achieved the highest performance for Model A ($R^2 = 0.25 \pm 0.02$, $f^2 = 0.28$, RMSE = 5.67, MAE = 4.59), followed by Model B ($R^2 = 0.20 \pm 0.01$, $f^2 = 0.25$, RMSE = 5.86, MAE = 4.76) and Model C ($R^2 = 0.16 \pm 0.02$, $f^2 = 0.16$, RMSE = 6.01, MAE = 4.90). These results corresponded to medium to large effect sizes, with the strongest effects observed in Model A.

BBAGs predict dementia status in wave 2: BBAG distributions differed by dementia status under multiple diagnostic definitions in Wave 2 (Supplementary S3 Fig. 1-D). Using the 10/66 and DSM-IV criteria, participants classified with dementia consistently showed accelerated aging across all model specifications. Effect sizes were large in Model A (10/66: $\delta = 0.79$; DSM-IV: $\delta = 0.82$; $p < 1\times10^{-15}$ for both), remained large in Model B (10/66: $\delta = 0.53$; DSM-IV: $\delta = 0.66$; $p < 1\times10^{-15}$), and were medium to large in Model C (10/66: $\delta = 0.50$; DSM-IV: $\delta = 0.55$; $p < 1\times10^{-15}$). Similar patterns were observed when applying CSID classification thresholds. Effect sizes decreased across progressively stricter cutoffs in Model A (Cut −1 vs −2: $\delta = 0.48$; Cut −2 vs −3: $\delta = 0.42$; Cut −1 vs −3: $\delta = 0.75$; all $p < 1\times10^{-15}$), corresponding to medium, medium, and large effects, respectively. In Models B and C, effects were small to medium (Model B: δ = 0.14, 0.31, and 0.43; Model C: δ = 0.23, 0.30, and 0.47).

Survival differences across diagnostic groups: Among healthy controls, all models showed highly significant differences between accelerated aging and delayed aging participants (Model A: log-rank $p < 1\times10^{-15}$; Model B: log-rank $p < 1\times10^{-15}$; Model C: log-rank $p < 1\times10^{-15}$). Similarly, in participants with dementia, significant effects were observed (Model A: log-rank $p = 5\times10^{-4}$; Model B: log-rank $p = 2\times10^{-6}$; Model C: log-rank $p < 1\times10^{-15}$) (Fig 2-B1).

Adjusted associations with mortality: These associations remained robust after adjustment for age, sex, and education, with each one standard deviation increase in BBAG conferring a 41–48% higher hazard of mortality (Model A: HR = 1.48, 95% CI 1.41–1.55; Model B: HR = 1.44, 95% CI 1.37–1.50; Model C: HR = 1.41, 95% CI 1.35–1.48; all $p < 1\times10^{-15}$) (Fig. 2-B2).

Unadjusted associations with mortality across diagnostic groups: In healthy controls, accelerated aging was consistently associated with increased mortality risk across all unadjusted models, with each one standard deviation increase in BBAG conferring a 37–40% higher hazard (Model A: HR = 1.37, 95% CI 1.29–1.45; Model B: HR = 1.40, 95% CI 1.32–1.48; Model C: HR = 1.38, 95% CI 1.31–1.46; all $p < 1\times10^{-15}$) (Supplementary S3 Fig. 1-E1). Among participants with dementia, accelerated aging was also associated with increased mortality risk across all models, with each one standard deviation increase in BBAG conferring a 25–45% higher hazard (Model A: HR = 1.39, 95% CI 1.24–1.56; Model B: HR = 1.45, 95% CI 1.31–1.62; Model C: HR = 1.25, 95% CI 1.12–1.39; all $p < 1\times10^{-4}$)(Supplementary S3 Fig. 1-E2).

Adjusted associations with mortality across diagnostic groups: In healthy controls, associations remained robust after adjustment for age, sex, and education, with each one standard deviation increase in BBAG conferring a 34–35% higher hazard of mortality across models (Model A: HR = 1.34, 95% CI 1.27–1.41; Model B: HR = 1.35, 95% CI 1.28–1.43; Model C: HR = 1.35, 95% CI 1.28–1.43; all $p < 1\times10^{-15}$) (Supplementary S3 Fig. 1-E1). Among participants with dementia, associations remained robust after adjustment for age, sex, and education, with each one standard deviation increase in BBAG conferring a 35–51% higher hazard of mortality across models (Model A: HR = 1.46, 95% CI 1.30–1.64; Model B: HR = 1.51, 95% CI 1.35–1.68; Model C: HR = 1.35, 95% CI 1.21–1.52; all $p < 1\times10^{-6}$)(Supplementary S3 Fig. 1-E2).

Comparison of mortality prediction between BBAGs and cardiometabolic and clinical indices: We performed bootstrap resampling (1,000 iterations) to compare hazard ratios between BBAGs and mortality-related cardiometabolic (hypertension, heart disease, angina and ICD-10 diagnoses), and clinical variables (Functional disability, Euro-D, ICD-10 and DSM-IV). We additionally compared BBAGs against a composite index combining all cardiometabolic, functional, depression and dementia measures. Continuous exposures were rescaled to the 0–1 range after outlier removal to ensure comparability with binary exposures. BBAGs consistently and significantly outperformed all other factors in predicting mortality (Supplementary S3 Fig. 1F).

3. Discussion

This study aimed to test whether biobehavioral clocks capture markers of pathological aging by predicting dementia status, mortality risk, and longitudinal aging trajectories. BBAGs accurately predicted chronological age across progressively stricter model specifications, demonstrating robustness even after excluding cognitive, functional, and mood predictors. Individuals with dementia consistently showed accelerated aging across diagnostic definitions and model variants. Accelerated aging was also associated with increased mortality risk, with effects persisting after adjustment for demographic covariates. Baseline BBAGs predicted future accelerated aging, supporting their temporal stability and prognostic value. Multiple additional analyses confirmed the consistency of the findings. Overall, these results suggest that biobehavioral aging may provide a scalable and ecologically grounded marker of accelerated aging linked to dementia and mortality in underrepresented populations.

Aging is distributed across multiple behavioral and health domains rather than driven solely by single factors such as cognition or functional impairment. This supports a whole-organism perspective of aging in which behavioral profiles integrate cumulative biological and environmental burden over time [5,19,52-54], consistent with multi-organ aging and frailty research[7,55-58]. The association between accelerated BBAGs and dementia, even after removing predictors related to diagnostic criteria, indicates that these measures capture latent pathological aging processes extending beyond classical cognitive decline. Unhealthy aging and dementia may reflect broader systemic dysregulation involving motor, sensory, cardiometabolic, systemic, and functional domains, in line with multidimensional models of neurodegeneration[59-63]. The association with survival further suggests that these measures capture distributed vulnerability across interacting physiological systems that shape long-term health trajectories. This interpretation is particularly relevant in Latin American populations, where adverse exposome profiles and elevated cardiometabolic risk contribute to heterogeneous aging patterns[10,35,64] and may amplify systemic pathways linking behavioral aging to dementia and mortality risk.

Biobehavioral age gaps provide a scalable approach to tracking aging trajectories in real-world functioning. Because BBAGs rely on behavioral, clinical, and sociodemographic variables that are routinely collected in population studies and healthcare settings, they may help identify risk for accelerated aging at scale. BBAGs can serve as proxies for systemic aging risk, bridging clinical outcomes with real-world functioning. This could be particularly relevant for detecting prodromal stages in settings where access to molecular biomarkers or neuroimaging remains limited[5,34,65]. BBAGs therefore offer a cost-effective and ecologically valid framework for monitoring aging, especially in under-resourced health systems. The longitudinal stability of BBAGs may reflect cumulative aging trajectories rather than transient events, suggesting that early deviations in biobehavioral profiles may contribute to downstream pathological aging cascades over time[1,2].

The present study has important strengths. It provides one of the first direct demonstrations linking biobehavioral clocks to both dementia and mortality, addressing a critical gap. The use of large multicountry population-based cohorts from Latin America improves generalizability to historically underrepresented populations and contributes to reducing global discrepancies in research[33,66]. The implementation of progressively restrictive models (excluding dementia-related predictors) and different diagnostic criteria provides a more rigorous test of circularity than the standard approaches in aging clocks. These tests help improve the internal validity of associations and strengthen the basis for future mechanistic research. The integration of harmonized behavioral, clinical, and socioeconomic variables offers a comprehensive representation of aging in real-world contexts[5]. Finally, advanced modeling approaches, including sequential feature selection, gradient boosting, and nested cross-validation, enhanced robustness and reduced overfitting. Similarly, the combination of cross-sectional, longitudinal, and survival and sensitivity analyses provides converging evidence across different inference levels.

This study also has important limitations. Although model restrictions reduced circularity, some predictors may still be indirectly related to dementia or health status. The observational design limits causal inference regarding mechanisms linking BBAGs to dementia and mortality. While geographically diverse, the sample is restricted to Latin American populations and requires validation in other global contexts. Residual measurement heterogeneity across sites may also have introduced variability despite harmonization procedures. In addition, the models did not incorporate biological or neuroimaging markers, calling for further comparative research with established molecular and brain aging clocks. Survival analyses may still reflect unmeasured confounding factors such as healthcare access or environmental exposures, and the follow-up period (3-5 years) may not fully capture long-term aging trajectories or late-stage outcomes.

In conclusion, this study suggests that biobehavioral aging is an ecologically grounded marker of pathological aging that predicts dementia, survival, and longitudinal aging trajectories independently of traditional diagnostic components. These findings support a whole-body and behaviorally embedded framework of aging and highlight the potential of scalable, noninvasive metrics to capture systemic vulnerability across diverse populations. Together, these results advance neurosyndemic models of organismal aging and open new avenues for early detection, risk stratification, and intervention, particularly in underrepresented settings.

4. Materials and methods

4.1. Cross-sectional participants

We used data from the Global 10/66 study cohort, a population-based study of older adults in low- and middle-income countries [36,37,67]. The cross-sectional sample included 12,693 participants (64.4% female; mean age = 74.7 years; s.d. = 7.19; range = 65–101; Dementia = 1,223) recruited across six Latin American and Caribbean countries (Cuba, Dominican Republic, Mexico, Peru, Puerto Rico, and Venezuela). Further demographic details are provided in Table 1.

4.2. Longitudinal participants

We used a longitudinal sample from the Global 10/66 study cohort, comprising 8,391 participants (65.9% females; mean age = 77.8 years; s.d. = 6.56; range = 66–101; Dementia = 1,198) recruited from the same six countries as the cross-sectional sample. Overall, 1,198 participants had dementia. Further demographic details are provided in Table 1. The baseline surveys were conducted between 2003 and 2005 for all sites apart from Puerto Rico, where data were collected from 2007. Response rates were excellent across all sites (72%-98%). A full follow-up was carried out 3 to 5 years after the baseline.

All participants were enrolled following informed signed consent. For individuals with dementia who lacked decision-making capacity, consent was obtained through written authorization from a relative. For participants who were illiterate, the study information and consent documents were read aloud, and verbal consent was obtained in the presence of a witness. The study protocol was approved by the local ethics committees in each participating country and by the ethics committee of the Institute of Psychiatry, King's College London.

4.3. Sample size calculation

We performed a power analysis using G*Power version 3.1.9.7 [68], employing an F-test for linear regressions. The necessary sample size for detecting small effect sizes (0.02 [69]) with a power of 0.99 was 921 participants ($\lambda = 18.42$; critical $F = 3.85$).

4.4. Predictors

We incorporated a set of ~~risk~~ protective and ~~protective~~ risk predictors for age estimation. Details regarding the harmonization procedures across study sites are available in Supplementary S1. Predictors were harmonized into a common numerical scale. (1) Dichotomous health conditions (hypertension, heart disease, stroke, transient ischemic attack (TIA), DSM-IV dementia [70], ICD-10 depressive episode [41], walking, medication consumption) were coded as binary variables 0/1. (2) Functional interference scales (vision, hearing, arthritis, respiratory symptoms, cardiac disorders, gastrointestinal impairment, syncope, paralysis, and skin-related problems) were coded on a four-point ordinal scale (0–3), reflecting both presence and activity limitation severity. (3) Composite scores [71] were derived by averaging related items: family dementia history (mother, up to four siblings), lifetime health burden score (malaria, tuberculosis, cysticercosis, sadness symptom), respiratory burden score (persistent cough, breathlessness interference), pain (frequency, severity, restriction), psychosocial functioning (emotional impact, concentration, social interaction, and friendship maintenance), and healthcare service utilization (dentist, private physician, traditional healer, and hospitalization in the past three months). (4) Motor signs (ataxia, bradykinesia) were retained as observed ordinal ratings. (5) Sociodemographic variables included sex (binary, self-reported) and educational attainment (five-level ordinal scale). (6) Continuous physiological and neuropsychological measures (blood pressure, blood pressure ratio, functional disability[42], mental distress[45], Neuropsychiatric score and Neuropsychiatric symptom severity[43], probable depression[44], global cognition (CSID)[39], immediate and delayed verbal memory (CERAD) [38], number of assets, illness-related burden) were retained in their original metric.

4.4.1. Factors for predicting age

Protective factors

Cognitive status: this factor included cognitive performance measures, including immediate and delayed verbal memory (CERAD) and a global cognitive score (CSID).

Behavioral and functional domains: this factor included physical activity, walking, and physical activity increase, reflecting functional capacity. Health service utilization and medication consumption were also included as proxies of healthcare engagement.

Sociodemographic and socioeconomic factors: the education level was divided into five categories: no formal education (none), elementary school, middle school, high school, and college university degrees. Asset index (number of assets) was included as a proxy of socioeconomic status.

Sex: sex assigned at birth was determined by self-report, with male sex analyzed as a protective factor [4]. No questions regarding gender identity aspects were asked in any participating country.

Risk factors

Health status: this factor included predictors related to general medical burden, comprising cardiometabolic conditions (hypertension, heart disease, stroke, transient ischemic attack, blood pressure, blood pressure ratio, angina), respiratory burden score (persistent cough, breathlessness, breath problems), musculoskeletal and neurological motor signs (arthritis, paralysis, ataxia, bradykinesia), gastrointestinal and dermatological complaints (gastrointestinal impairment, skin-related problems), and dementia diagnostic indicators (DSM-IV and ICD-10 criteria). Additionally, sensory impairments (vision and hearing problems), pain, lifetime health burden score, fainting episodes, functional disability (WHO-DAS II), family dementia history, and illness-related burden were included as indicators of overall health deterioration.

Neuropsychiatric and psychological burden: this factor included neuropsychiatric symptom burden and symptom severity derived from the Neuropsychiatric Inventory (NPI), probable depression measured using the Euro-D scale, mental distress assessed with the Self-Reporting Questionnaire (SRQ), and psychosocial functioning (composite score of emotional regulation, sustained concentration, managing unfamiliar social interactions, and maintaining friendships; each component ranged from 0 to 4).

4.5. Model definitions

To examine the robustness of the predictive framework and to evaluate the influence of specific predictor groups, we trained three model configurations that differed in the set of predictors included.

Model A (full model) incorporated the full set of predictors available in the study, allowing the algorithm to leverage all biobehavioral information collected from participants.

Model B (cognitive-negative) excluded predictors that are directly used within the diagnostic algorithms defining the clinical groups in the Global 10/66 dataset (cognitive score, CERAD immediate recall, CERAD delayed recall, DSM-IV, and ICD-10), thereby reducing potential circularity between model inputs and diagnostic classifications.

Model C (cognitive-mood-functional-negative) applied an even more restrictive specification by additionally removing depressive and functional ability variables (disability status, Euro-D scale, SRQ, and emotional disability), which are closely related to clinical dementia assessments. This step allowed us to evaluate whether the predictive signal of the model could still be captured using non-depressive and non-functional health indicators alone, providing a stricter test of the independence and robustness of the biobehavioral aging signal.

4.6. Statistical analysis

4.6.1. Sequential feature selection (SFS)

Sequential feature selection algorithms[46] represent a group of greedy optimization approaches that progressively reduce an original feature set of dimension d to a smaller subset containing k variables~~, where k is smaller than d~~. Their purpose is to automatically determine which predictors contribute most to the predictive task. Eliminating redundant or noisy features can make the model computationally more efficient and may also improve its generalization performance [46,72].

4.6.2. Predictor-based age estimation

We utilized histogram-based Gradient Boosting [47,48] regression models to predict the age of participants using demographic and behavioral predictors. A nested cross-validation was used to predict chronological age and optimize model performance. In each outer-loop iteration, one fold was held out for testing, while the remaining folds were used in an inner loop to tune hyperparameters through Bayesian optimization [49]. The optimized parameters included learning rate (0.01–0.1), maximum tree depth (3–7), and the number of iterations (50–200). After identifying the optimal configuration, the model was retrained on the complete inner training set and evaluated on the outer test fold. For each iteration, we computed feature importance, R^2, Cohen's f^2 [73], Mean Directional Error (MDE), and Root Mean Squared Error (RMSE). Feature importance was estimated using permutation importance.

This procedure was repeated across all outer folds, ensuring that each fold served once as the test set. Nested cross-validation explicitly separates hyperparameter optimization (inner loop = 5) from unbiased performance estimation (outer loop = 10), preventing data leakage and providing robust estimates of model generalization [74,75]. Unlike single train–test splits, which may produce unstable estimates, nested cross-validation generates a distribution of performance metrics while efficiently leveraging the available data. Consequently, it is widely considered a gold standard for model evaluation in high-dimensional settings [76].

4.6.3. BBAGs estimation

BBAGs [4] were calculated by subtracting the chronological age from the predicted age using the Gradient Boosting model. This yields a normalized individual metric that shows whether each participant appears older (BBAG > 0) or younger (BBAG < 0) compared to their chronological age. Chronological age was regressed from all estimated BBAGs to correct for regression-to-the-mean bias [77,78]. The residuals from this regression defined the adjusted BBAGs. The regression coefficients for adjusting the BBAGs were fitted using the training set and then applied to adjust the BBAGs in the test set.

4.6.4 Survival analysis

To examine the association between BBAGs and mortality, we employed Cox proportional hazards regression models [50]. The time-to-event variable was defined as the time from baseline assessment to death, with participants who did not experience the event being censored at the date of last follow-up. Hazard ratios (HR) and their 95% confidence intervals (CI) were reported. Survival functions were estimated using the Kaplan-Meier method [51] and compared across groups using the log-rank test. All analyses were conducted using Python, with a significance threshold set at $p < 0.05$.

4.6.5. Linear Regression

We used linear regression[79] models to assess the association between BBAGs from different waves. The mathematical representation of the linear regression model is:

$$Y = B_0 + B_1 x + \epsilon$$

where Y is the outcome, variable, B_0 is the intercept, $B = [B_1, B_2, \dots, B_n]$ is the coefficients vector of the independent variables matrix $X = [x_1, x_2, \dots, x_n]$ and ϵ represents the error term. The coefficients $B_1, B_2, \dots, B_n$ are determined using the least squares method, which aims to minimize the sum of the squared differences (errors) between the observed values and the values predicted by the model. The formulas to calculate the coefficients are:

$$B = (X^T X)^{-1} X^T Y$$

where X^T denotes the transpose of matrix X.

4.7. Meta-analysis

Meta-analyses were used to explore the consistency of the association between BBAGs and mortality across countries and models. Hazard ratios were calculated for each country. Meta-analyses employed both random-effects and common-effects models to estimate pooled effect sizes. I^2 values exceeding 50% indicated substantial heterogeneity. The Restricted Maximum Likelihood (REML) estimator calculated between-country variance.

4.8. Sensitivity analysis

Several sensitivity analyses were performed to evaluate the stability and generalizability of the findings. Chronological age prediction models were re-estimated using only healthy participants and using data from the follow-up wave (wave 2) to assess model robustness across samples and time points. Additionally, the discriminative validity of BBAG was examined by comparing its distribution across dementia status. We also performed a meta-analysis restricted to healthy participants to assess whether the association between accelerated aging and mortality risk was preserved in the absence of clinical diagnoses.

Survival differences associated with BBAG were further evaluated within diagnostic subgroups using Kaplan–Meier estimators and log-rank tests. Cox proportional hazards models were re-estimated both unadjusted and after adjustment for demographic covariates, including age, sex, and education, to control for potential confounding influences. Stratified analyses across diagnostic groups were conducted to assess whether the associations between BBAG and mortality differed according to clinical status.

Finally, we implemented a bootstrap-based procedure (1,000 iterations) to statistically compare hazard ratios across a) BBAGs, b) cardiometabolic conditions (hypertension, heart disease, angina and ICD-10 diagnoses), and c) clinical variables (functionality, depression and dementia indices), including a composite index aggregating all exposures. To ensure comparability across measures, continuous variables were rescaled to the 0–1 interval following outlier removal. Statistical significance was evaluated using permutation tests with 5,000 iterations and Bonferroni correction.

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Data sharing statement

The data used in this study are part of the 10/66 Dementia Research Group's research and are openly accessible upon request. Access is granted after meeting the requirements for data usage outlined by the Alzheimer's Disease Data Initiative (ADDI). Researchers can access the data through the ADDI platform at https://fair.addi.ad-datainitiative.org/#/data/collections/1066-dementia-research-group.

Code availability

Analysis codes are freely available on GitHub (https://github.com/euroladbrainlat/Biobehavioral-clocks-predict-dementia-and-future-survival).

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Supplementary Data

Supplementary S1. Measurements

Familial dementia history. Self-report question assessing whether any close relatives (parents or up to four siblings) had experienced severe memory problems interfering with daily functioning; used as a proxy for family history of dementia.
Sex. Binary variable coded as 0 = female and 1 = male.
Education. Harmonized into five ordinal categories reflecting highest completed level: 0 = no formal education, 1 = elementary school, 2 = middle school, 3 = high school, and 4 = college/university.
Functional disability. Assessed using the total WHO-DAS II score (12 items), capturing the extent to which health-related limitations affect daily functioning.
Asset index. Proxy for socioeconomic status based on the number of household assets, with higher values indicating greater material resources.
Ataxia. Presence of gait unsteadiness observed during a 10-meter walk.
Bradykinesia. Slowed movement evaluated through clinical observation while walking.
Hypertension. Self-reported physician diagnosis of high blood pressure (0 = no, 1 = yes).
Heart disease. Self-reported physician diagnosis of a cardiac condition (0 = no, 1 = yes).
Visual impairment. Self-reported vision problems categorized by interference with daily activities (0 = none, 1 = present without interference, 2 = mild interference, 3 = severe interference).

Hearing impairment. Classified analogously based on self-reported hearing difficulties and their impact on daily functioning (0–3 scale).
General cognition. Total score from the CSID participant cognitive test.
Immediate verbal memory. CERAD 10-word list immediate recall total score (range 0–30).
Delayed verbal memory. CERAD delayed recall score.
DSM-IV dementia. Dementia status determined according to DSM-IV criteria.
Depressive episode. Presence of an ICD-10 depressive episode.
Probable depression. EURO-D score ≥4 used as threshold.
Blood pressure. Pulse pressure calculated as the difference between mean systolic and mean diastolic blood pressure across three measurements.
Blood pressure ratio. Ratio between mean systolic and diastolic blood pressure across three measurements.
Stroke. History of sudden-onset neurological symptoms (e.g., unilateral weakness, speech loss, or blindness) lasting at least two days and requiring medical attention (0 = no, 1 = yes).
Transient ischemic attack. Similar neurological symptoms resolving within 24 hours of onset (0 = no, 1 = yes).
Lifetime health burden score. Composite measure integrating duration of recurrent symptoms, history and recency of infectious diseases (malaria, tuberculosis, cysticercosis), and prior episodes of prolonged depressive symptoms.
Mental distress. Total score from the Self-Reporting Questionnaire (SRQ).
Neuropsychiatric score. Neuropsychiatric Inventory (NPI) total distress score.
Neuropsychiatric symptom severity. NPI total severity score.
Arthritis. Self-reported arthritis or rheumatism categorized by degree of interference with daily activities (0–3 scale).
Persistent cough. Chronic cough classified according to its impact on daily functioning (0–3 scale).
Breathlessness. Shortness of breath or asthma symptoms categorized by interference with daily activities (0–3 scale).
Respiratory burden score. Composite index combining persistent cough and breathlessness indicators.
Angina. Self-reported cardiac symptoms categorized according to their interference with daily functioning (0–3 scale).
Gastrointestinal impairment. Self-reported gastrointestinal problems classified by level of interference (0–3 scale).
Fainting episodes. Episodes of syncope categorized based on their impact on daily activities (0–3 scale).
Paralysis. Self-reported paralysis or limb weakness categorized by degree of interference (0–3 scale).
Illness-related burden. Number of limiting illnesses recoded as 0 = none, 1 = one to two, and 2 = three or more conditions.
Skin-related problems. Skin conditions (e.g., ulcers or severe burns) categorized by interference with daily functioning (0–3 scale).
Pain. Composite score combining frequency, severity, and functional limitation due to pain in the past month.
Psychosocial functioning. Composite measure reflecting emotional distress, concentration difficulties, social interaction challenges, and problems maintaining relationships.
Physical activity. Self-reported activity level (work and leisure), reverse-coded so that higher values indicate higher activity levels.
Walking. Indicator of having walked at least 0.5 km continuously during the previous month (0 = no, 1 = yes).
Physical activity increase. Change in exercise level relative to 10 years earlier (1 = less, 2 = same, 3 = more).
Health service utilization. Composite score reflecting use of healthcare services (dentist, private doctor, traditional healer, or hospitalization) in the previous three months.

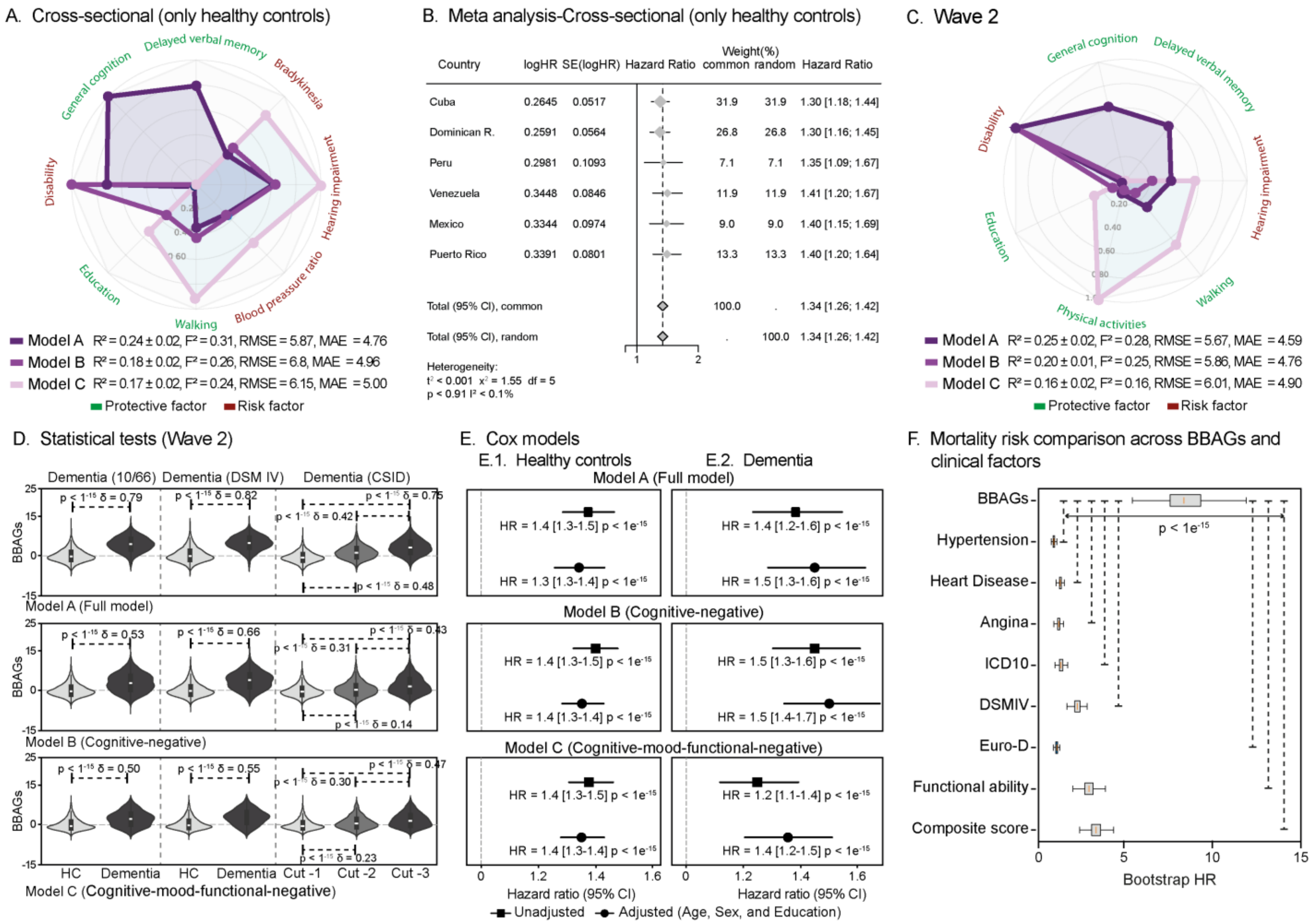
Medication consumption. Indicator of medication use during the previous three months (0 = no, 1 = yes).

Supplementary S2. Tables

Supplementary Table 1. Feature importance.

Factors	Model A	Model B	Model C
General cognition	0.08762	-	-
Delayed verbal memory	0.05051	-	-
Hearing impairment	0.03557	0.04217	0.04701
Functional disability	0.03452	0.11141	
Bradykinesia	0.02030	0.03104	0.03789
Walking	0.01933	0.03450	0.05287
Blood pressure	0.01718	0.02424	0.02584
Probable Depression	0.01066	0.01313	
Physical activity	0.00864	0.01825	0.05085
Psychosocial functioning	0.00818	0.01141	-
Asset index	0.00469	-	-
Hypertension	0.00456	0.00893	0.01045
Heart disease	0.00329	0.00393	0.00501
Physical activity increase	0.00323	-	0.00505
Ataxia	0.00312	0.00839	0.00875
Stroke	0.00268	0.00107	-
Visual impairment	0.00242	0.00228	0.00285
Education	0.00224	0.02788	0.02681
Pain	0.00185	0.00906	0.00836
Arthritis	0.00168	-	0.00250
Lifetime health burden score	0.00134	-	0.00096
Sex	0.00129	0.00355	0.00160
Medication consumption	0.00106	0.00209	0.00259
Paralysis	0.00097	-	-
Blood pressure ratio	0.00078	-	0.00436
Fainting episodes	0.00074	-	-
Neuropsychiatric symptom severity	0.00070	-	-
Breathlessness	0.00047	0.00234	0.00318
Depressive episode	0.00035	-	-
Mental Distress	0.00028	-	-
Angina	0.00022	-	0.00072
Familial Memory Impairment	0.00008	-	0.00000
Gastrointestinal impairment	0.00006	0.00140	-
Respiratory burden score	0.00001	0.00019	0.00007
Skin-related problems	0.00004	-	0.00006
DSM-IV dementia	0.00005	-	-
Transient ischemic attack	-	0.00222	-
Persistent cough	-	0.00034	0.00014
Illness-related burden	-	-	0.00046

Supplementary S3. Figures



Supplementary Figure 1. Sensitivity analysis. (A) Healthy control models. Model performance and feature contributions across three model configurations using the healthy control participants: Model A (full model), Model B (cognitive-negative), and Model C (cognitive-mood-functional-negative). Radar plots display the relative contribution of predictors grouped into cognitive, behavioral, and clinical domains, distinguishing protective (green) and risk (red) factors. Model performance metrics (R^2, f^2, RMSE, and MAE) are reported for each configuration. (B) **Meta-analysis across countries in healthy controls**. Biobehavioral age gap (BBAG) and mortality risk across countries. Forest plots show hazard ratios (HRs) and 95% confidence intervals for each country and pooled estimates using both common- and random-effects models. **(C) Follow-up performance.** Model performance and feature contributions across three model configurations using wave 2 data: Model A (full model), Model B (cognitive-negative), and Model C (cognitive-mood-functional-negative). **(D) Statistical tests.** Statistical comparisons (two-sided tests) indicate significant differences across diagnostic categories. **(E) Cox models.** Cox proportional hazards models showing the association between BBAGs and mortality risk. Hazard ratios (HR) are reported for unadjusted and adjusted models (age, sex, and education), demonstrating a robust association between higher BBAGs (accelerated aging) and increased mortality risk across all model configurations. (**F) Mortality risk comparison across BBAGs and clinical factors.** Bootstrap distributions of hazard ratios (HRs) for mortality across BBAGs, cardiometabolic conditions, and functional, depression and dementia indices, including a composite score. BBAGs showed significantly stronger associations with mortality than all other predictors (permutation tests, 5,000 iterations, Bonferroni-corrected; $p < 1 \times 10^{-15}$).